

Theory Based Inference I

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Stat 141 | Fall 2026

Friday, Week 11

Attendance sign-in:

https://tinyurl.com/stat141-signin

Goals for today

- Review how to conduct inference
- Apply the Central Limit Theorem to data from a Binomial distribution
- Practice "theory-based inference" for one proportion

Review of Inference

What's Inference Again?

Statistical Inference: Estimating or testing claims about a **parameter** in a **population**, using a **statistic** in a **sample**.

- **Ex:** To understand proportion of Reedies who are vegetarians, we survey 40 students. 16 of the 40 are vegetarians.
  - **Population:** All Reedies
  - **Parameter:** p = Proportion of all Reedies who are vegetarians
  - **Sample:** $n = 40$ Reedies we surveyed
  - **Statistic:** $\hat{p} = 16/40 = 0.4$
- We have seen two types of inference procedures:
  1. **Confidence Intervals**
  2. **Hypothesis Testing**
- Let's review each…

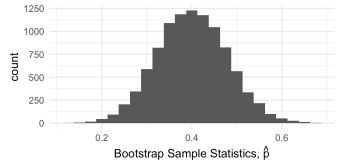
Confidence Intervals

Confidence Intervals: plausible ranges of values for a parameter

Ex: What proportion of Reedies are vegetarians?

- **Population:** All Reedies
- **Parameter:** p = Proportion of Reedies who are vegetarians
- **Sample:** $n = 40$ students, of which 16 were vegetarian.
- **Statistic:** $\hat{p} = 16/40 = 0.4$

Until now, we have learned to construct confidence intervals using **Bootstrap Distributions**:



- **Bootstrap distribution:** Approximation of sampling distribution
- Sample n observations with replacement from the sample
- For each "bootstrap sample", calculate/plot the statistic
- **90% confidence interval:** Middle 90% is $(0.275,\ 0.525)$

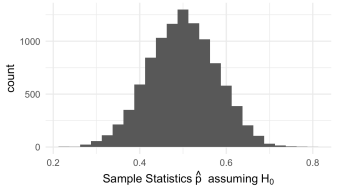
Hypothesis Testing

Hypothesis Tests: formal procedure for testing claims about a parameter

Ex: Are *less than* 50% of Reedies vegetarians?

- **Parameter:** p = Proportion of Reedies who are vegetarians
- **Statistic:** $\hat{p} = 16/40 = 0.4$
- **State Hypotheses:** $H_0 : p = 0.5$; $H_a : p < 0.5$

We need to construct the **Null Distribution:**



- **Null distribution:** Sampling distribution assuming H_0
- Many times, sample n observations from population assumed under H_0
- For each "sample", calculate/plot the statistic
- **P-Value:** Area under curve to the left of $\hat{p}$ is 0.07
- **Compare to Significance Level:** Using $\alpha = 0.05$, we would *fail to reject* H_0.

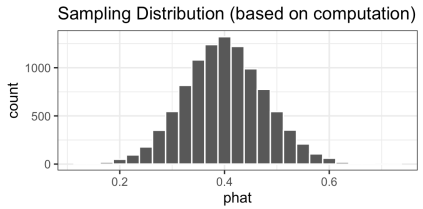
What's Similar?

- **Both inference procedures require a sampling distribution of a statistic:**
  - Confidence Intervals: Approximate with our bootstrap distribution (based on our sample)
  - Hypothesis Testing: Simulate a sampling distribution assuming a population under H_0
- We've had to use simulation in R to find these "distributions of statistics"
- **Could we utilize probability theory to do this work more efficiently? What did people use before computers?**

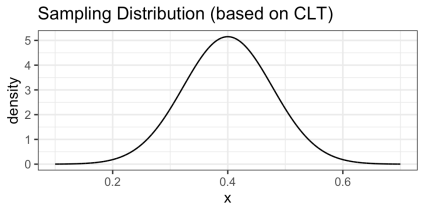
Theory-Based Inference

- Could we do this work more efficiently? **Yes! Use the Central Limit Theorem (CLT)!**
- Last lecture, we found:
  - Many sample statistics can be written as a **"sample mean"**
  - Sample means follow **Normal distributions**, with **formulas** for the mean and SD
  - We can **use the CLT** to characterize the distribution of a statistic!

Instead of…



… we will instead use:



Inference for a Single Proportion

Central Limit Theorem (CLT)

Theorem: Central Limit Theorem

Suppose a simple random sample of size n is drawn from a population with finite mean μ and finite standard deviation σ. Let $\bar{x}$ be the sample mean. When n is large, then approximately

$$\bar{x} \sim Normal\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

Goal: Can we write a sample *proportion* $\hat{p}$ as a sample *mean*?

Sample Proportion as a Sample Mean

- Reminder: Let $X \sim \text{Binomial}(n, p)$ distribution
  - X is number of "successes" in n independent trials, each trial has probability of success, p
  - $E(X) = np$ and $SD(X) = \sqrt{np(1-p)}$
- **Ex:** Polling (a SRS of) n Reedies about vegetarianism

- **Key Trick 1:** Define
  - $X_1 = 1$ if Person 1 is vegetarian, 0 if not
  - $\vdots$
  - $X_n = 1$ if Person n is vegetarian, 0 if not
- Note each $X_i \sim \text{Bernoulli}(p)$

- **Key Trick 2:** The sample proportion, $\hat{p}$ is a sample mean of these X_i's!

$$\begin{aligned}\hat{p} &= \frac{\text{Number of vegetarians}}{n} \\ &= \frac{X_1 + X_2 + \cdots + X_n}{n} \\ &= \bar{x}\end{aligned}$$

Proportions and the CLT

Theorem: Central Limit Theorem

Suppose a simple random sample of size n is drawn from a population with finite mean μ and finite standard deviation σ. Let $\bar{x}$ be the sample mean. When n is large, then approximately

$$\bar{x} \sim Normal\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

Bernoulli Random Variables: If $X_i \sim \text{Bernoulli}(p)$, then

$$\mu = E(X_i) = p \quad \text{and} \quad \sigma = SD(X) = \sqrt{p(1-p)}$$

- $\hat{p} = \bar{x}$
- Plugging this in to CLT, $\hat{p} \sim \text{Normal}\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$

Theory-Based Sampling Distribution for One Sample Proportion

Theorem: Theory-Based Sampling Distribution for One Sample Proportion

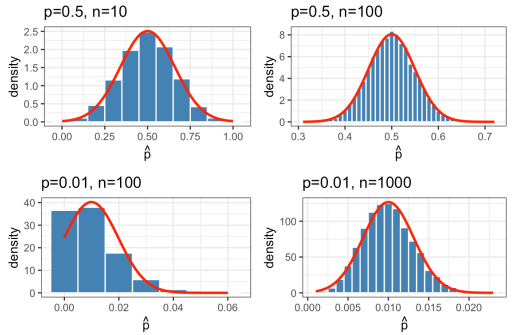
Given a SRS of size n drawn from a population with true proportion p, when n is large the sample proportion $\hat{p}$ is distributed as:

$$\hat{p} \sim \text{Normal}\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$$

- Traditionally, n is considered "large enough" when there are:
  - 10 or more "successes", and
  - 10 or more "failures"
- Mathematically,
  - $n\hat{p} \geq 10$, and
  - $n(1-\hat{p}) \geq 10$

Word of Caution: CLT requires "large" n

Theory-Based Sampling Distribution: $\hat{p} \sim \text{Normal}\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$



- **"Large Enough n"**: 10 or more "successes" *and* "failures"
- **Q:** When does the (red) theory-based curve best/worst match the true sampling distribution? Why?

Confidence Intervals and Hypothesis Tests

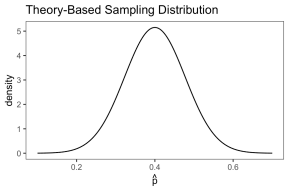
Confidence Intervals (again)

Ex: What proportion of Reedies are vegetarians?

- **Population:** All Reedies
- **Parameter:** p = Proportion of Reedies who are vegetarians
- **Sample:** $n = 40$ students, of which 16 were vegetarian.
- **Statistic:** $\hat{p} = 16/40 = 0.4$

Sampling Distribution:

- *Theoretical* sampling distribution:
  $\hat{p} \sim \text{Normal}(p, \sqrt{p(1-p)/n})$
- For *our best guess*, **substitute $\hat{p} = 0.4$ for p**:
  $\hat{p} \sim \text{Normal}(0.4, \sqrt{0.4(1-0.4)/40})$



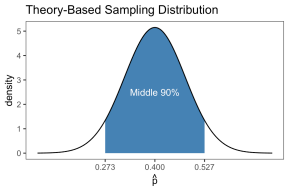
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  $\hat{p} \sim \text{Normal}(p, \sqrt{p(1-p)/n})$
- For *our best guess*, substitute $\hat{p} = 0.4$ for p:
  $\hat{p} \sim \text{Normal}(0.4, \sqrt{0.4(1-0.4)/40})$
- **Quantiles** of this distribution give us confidence intervals, ex. $(0.273,\ 0.527)$



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Confidence Intervals: Process Check

When making (percentile) confidence intervals using **computational** methods, we:

1. Specify our population, and parameter
2. Collect a sample, and calculate the statistic
3. Create a bootstrap distribution to approximate the sampling distribution
4. Use quantiles of the approximated sampling distribution to make a confidence interval

Q: Which steps above are the **same** and which are **different** with our new "theory-based" way?

- **Only #3 is different!**
- "Old Way": Collect bootstrap samples
- "Theory-based Way": Use CLT, which gives a "theoretical" sampling distribution of

$$\hat{p} \sim \text{Normal}(\hat{p}, \sqrt{\hat{p}(1-\hat{p})/n})$$

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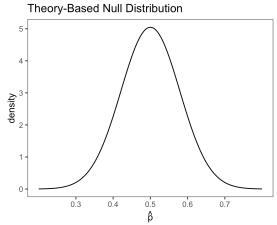
Hypothesis Testing (again)

Ex: Are *less than* 50% of Reedies vegetarians?

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- **State Hypotheses:** $H_0 : p = 0.5$; $H_a : p < 0.5$

Null Distribution:

- *Theoretical* sampling distribution:
  $\hat{p} \sim \text{Normal}(p, \sqrt{p(1-p)/n})$
- For *null distribution*, substitute "null value" $p = 0.5$:
  $\hat{p} \sim \text{Normal}(0.5, \sqrt{0.5(1-0.5)/40})$



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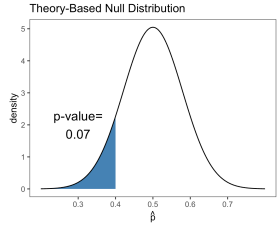
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- For *null distribution*, substitute "null value" $p = 0.5$:
  $\hat{p} \sim \text{Normal}(0.5, \sqrt{0.5(1-0.5)/40})$
- **P-Value:** In this case, area under curve to the left of $\hat{p} = 0.4$. Here, $\textbf{P-Value} = 0.07$.
- Compare the P-Value to your significance level (α).



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Hypothesis Tests: Process Check

When conducting hypothesis tests, we:

1. Specify our population, and parameter
2. State null (H_0) and alternative (H_a) hypotheses, e.g.,

$$H_0 : p = p_0 \qquad H_a : p \neq p_0 \text{ or } p < p_0 \text{ or } p > p_0$$

3. Collect a sample, and calculate the statistic
4. Create a null distribution
5. Use "areas/proportions" of the null distribution distribution to calculate the p-value
6. Compare p-value to our rejection threshold α, and make a conclusion

- **Q:** Which steps in the above are the **same** in both the "old" way of doing hypothesis tests, and this new "theory-based" way? Which steps are **different**?
- **Only #4 is different!** (again, the distribution we use comes from theory instead of simulation)

Activity

Activity

In a SRS of 60 Reedies, 45 have watched the *Lord of the Rings* trilogy. Create a **80% confidence interval** for the true proportion of Reedies who have watched *LOTR* using theory-based inference.

1. Identify the population, sample, parameter, statistic.
2. Sketch and write down the assumed sampling distribution of the statistic
3. Write R code that would calculate the confidence interval
4. Suppose you get [0.68, 0.82]. Interpret this confidence interval in context.

In a SRS of 30 birds in the Reed canyon, 20 are crows. Do we have evidence that more than 60% of birds in the Reed canyon are crows? Conduct a **hypothesis test** using theory-based inference.

1. Identify the population, sample, parameter, statistic, and **hypotheses**.
2. Sketch and write down the null distribution
3. Write R code that would calculate the p-value.
4. Suppose your p-value is 0.23. Use this p-value to make a conclusion with $\alpha = 0.10$.

05:00

Recap: Theory-Based Inference

CLT-Based Sampling Distribution: $\hat{p} \sim \text{Normal}\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$

- The general steps we take are the same as before, but we **now use the CLT instead of simulation to find sampling distributions**.

Confidence Intervals	**Hypothesis Testing**
Sampling Distribution of p	**Null Distribution**
(Plug in statistic, $\hat{p}$)	**(Plug in null value, p_0)**
$\text{Normal}\left(\hat{p}, \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right)$	$\text{Normal}\left(p_0, \sqrt{\frac{p_0(1-p_0)}{n}}\right)$